

OS LAB MANUAL

(CS23431)

Lab:3

Roll No:230701229

EX.NO:6D

ROUND ROBIN SCHEDULING

Aim: To implement the Round Robin (RR) scheduling technique

Program:

```
#include <stdio.h>

struct Process {

    int id;

    int burst_time;
    int remaining_burst_time;

    int waiting_time;

    int turnaround_time;

};

int main() {

    int n, quantum, t = 0;

    struct Process p[10];

    int total_waiting_time = 0, total_turnaround_time = 0;

    printf("Enter the number of processes: ");

    scanf("%d", &n);


    printf("Enter the time quantum: ");

    scanf("%d", &quantum);

    printf("Enter the burst time of the processes:\n");
```

```

for (int i = 0; i < n; i++) {
    printf("Process %d - Burst Time: ", i + 1);
    scanf("%d", &p[i].burst_time);
    p[i].remaining_burst_time = p[i].burst_time;
    p[i].waiting_time = 0;
    p[i].turnaround_time = 0;
    p[i].id = i + 1;
}
while (1) {
    int done = 1;
    for (int i = 0; i < n; i++) {
        if (p[i].remaining_burst_time > 0) {
            done = 0;
            if (p[i].remaining_burst_time > quantum) {
                t += quantum;
                p[i].remaining_burst_time -= quantum;
            } else {
t += p[i].remaining_burst_time;
                p[i].waiting_time = t - p[i].burst_time;
                p[i].turnaround_time = t;
                p[i].remaining_burst_time = 0;
            }
        }
    }
}

printf("\nProcess\tBurst Time\tWaiting Time\tTurn Around
Time\n");

for (int i = 0; i < n; i++) {
    printf("%d\t%d\t\t%d\t\t%d\n", p[i].id, p[i].burst_time,
p[i].waiting_time, p[i].turnaround_time);
}

```

```

    printf("\nAverage waiting time is: %.2f",
(float)total_waiting_time / n);

    printf("\nAverage Turn Around Time is: %.2f\n",
(float)total_turnaround_time / n);

    return 0;
}

```

INPUT:

```

Enter the number of processes: 4
Enter the time quantum: 4
Enter the burst time of the processes
Process 1 - Burst Time: 6
Process 2 - Burst Time: 8
Process 3 - Burst Time: 5
Process 4 - Burst Time: 4

```

OUTPUT:

```

Enter the number of processes: 4
Enter the time quantum: 4
Enter the burst time of the processes:
Process 1 - Burst Time: 6
Process 2 - Burst Time: 8
Process 3 - Burst Time: 5
Process 4 - Burst Time: 4

Process Burst Time      Waiting Time      Turn Around Time
1         6             12                18
2         8             14                22
3         5             18                23
4         4             12                16

Average waiting time is: 14.00
Average Turn Around Time is: 19.75

```


